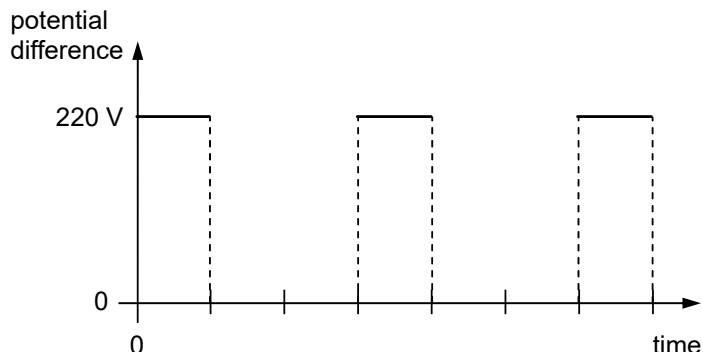


28. An alternating current is supplied to a rectifier. The output from the rectifier has the shape shown in the diagram, with a maximum value of 220 V.



This output is used as the supply to a resistor of resistance 60.0 Ω .

What are the r.m.s. values of the current and the power to the resistor?

	r.m.s. current / A	power / W
A	1.22	89.6
B	1.83	202
C	2.12	269
D	2.59	403

Answer: C

$$\text{r.m.s. voltage} = \sqrt{\frac{220^2 \times \frac{1}{3}T}{T}} = 127.02 \text{ V}$$

$$\text{r.m.s. current} = \frac{127.02}{60} = 2.117 \text{ A}$$

$$\text{power} = I_{\text{r.m.s.}}^2 R = 2.117^2 \times 60 = 269 \text{ W}$$